

WGUPS Routing Program

Evidence Screenshots

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The following figures show the program output at the times required by the scenario.

Status Check at 8:35 a.m.

Figure 1. Package status at 8:35 a.m. All forty packages are visible on their assigned trucks. Packages 6, 9, 25, 28, and 32 appear in “delayed” status. Packages 2, 7, 14, 15, 16, 29, 33, and 34 are already delivered. Truck 1 is en route. Truck 2 packages remain at the hub because that truck does not depart until 9:05 a.m. Truck 3 packages also remain at the hub because that truck has not yet departed.

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bradwhite /c950 P-mastan just run
WGUPS Routing Program
Total mileage for all trucks: 89.3 miles
Enter a time (e.g. '8:35 AM', '12:03 PM', '1:12 PM') or 'all' for final status, or 'q' to quit: 8:35 AM
Status at 8:35 AM

Truck 1 16 packages (en route (departed 8:00 AM))
Package 2 delivered at 8:34 AM Deadline: EOD 2530 S 500 E
Package 7 delivered at 8:29 AM Deadline: EOD 1330 2100 S
Package 12 en route Deadline: EOD 3575 W Valley Central Station bus Loop
Package 13 en route Deadline: 10:30:00 2010 W 500 S
Package 14 delivered at 8:18 AM Deadline: 10:30:00 4300 S 1300 E
Package 15 delivered at 8:11 AM Deadline: 09:00:00 4580 S 2300 E
Package 16 delivered at 8:11 AM Deadline: 10:30:00 4580 S 2300 E
Package 19 en route Deadline: EOD 177 W Price Ave
Package 20 en route Deadline: 10:30:00 3595 Main St
Package 21 en route Deadline: EOD 3595 Main St
Package 29 delivered at 8:29 AM Deadline: 10:30:00 1330 2100 S
Package 31 en route Deadline: 10:30:00 3365 S 900 W
Package 33 delivered at 8:34 AM Deadline: EOD 2530 S 500 E
Package 34 delivered at 8:11 AM Deadline: 10:30:00 4580 S 2300 E
Package 39 en route Deadline: EOD 2010 W 500 S
Package 40 en route Deadline: 10:30:00 380 W 2880 S

Truck 2 16 packages (at hub)
Package 1 at hub Deadline: 10:30:00 195 W Oakland Ave
Package 3 at hub Deadline: EOD 233 Canyon Rd
Package 4 at hub Deadline: EOD 380 W 2880 S
Package 6 delayed Deadline: 10:30:00 3060 Lester St
Package 10 at hub Deadline: EOD 600 E 900 South
Package 11 at hub Deadline: EOD 2600 Taylorsville Blvd
Package 17 at hub Deadline: EOD 3148 S 1100 W
Package 18 at hub Deadline: EOD 1488 4800 S
Package 23 at hub Deadline: EOD 5100 South 2700 West
Package 25 delayed Deadline: 10:30:00 5383 S 900 East #104
Package 28 delayed Deadline: EOD 2835 Main St
Package 30 at hub Deadline: 10:30:00 300 State St
Package 32 delayed Deadline: EOD 3365 S 900 W
Package 36 at hub Deadline: EOD 2300 Parkway Blvd
Package 37 at hub Deadline: 10:30:00 300 State St
Package 38 at hub Deadline: EOD 300 State St

Truck 3 8 packages (at hub)
Package 5 at hub Deadline: EOD 300 State St
Package 8 at hub Deadline: EOD 300 State St
Package 9 delayed Deadline: EOD 410 S State St
Package 22 at hub Deadline: EOD 6351 South 900 East
Package 24 at hub Deadline: EOD 5025 State St
Package 26 at hub Deadline: EOD 5383 S 900 East #104
Package 27 at hub Deadline: EOD 1060 Dalton Ave S
Package 35 at hub Deadline: EOD 1060 Dalton Ave S

Summary
delivered: 8
en route: 8
at hub: 19
delayed: 5

Enter a time (e.g. '8:35 AM', '12:03 PM', '1:12 PM') or 'all' for final status, or 'q' to quit: 
```

Status Check at 9:35 a.m.

Figure 2. Package status at 9:35 a.m. Package 9 is still “delayed” because its address is not corrected until 10:20 a.m. Packages 6, 25, 28, and 32 are now “en route” or already delivered because Truck 2 departed at 9:05 a.m. Truck 3’s packages remain “at hub” pending the 10:30 a.m. departure.

```
Enter a time (e.g. '8:35 AM', '12:03 PM', '1:12 PM') or 'all' for final status, or 'q' to quit: 9:35 AM

Status at 9:35 AM

Truck 1 16 packages (en route (departed 8:00 AM))
Package 2 delivered at 8:34 AM      Deadline: EOD      2530 S 500 E
Package 7 delivered at 8:29 AM      Deadline: EOD      1330 2100 S
Package 12 delivered at 9:29 AM      Deadline: EOD      3575 W Valley Central Station bus Loop
Package 13 delivered at 9:05 AM      Deadline: 10:30:00 2010 W 500 S
Package 14 delivered at 8:18 AM      Deadline: 10:30:00 4300 S 1300 E
Package 15 delivered at 8:11 AM      Deadline: 09:00:00 4580 S 2300 E
Package 16 delivered at 8:11 AM      Deadline: 10:30:00 4580 S 2300 E
Package 19 delivered at 9:34 AM      Deadline: EOD      177 W Price Ave
Package 20 en route                 Deadline: 10:30:00 3595 Main St
Package 21 en route                 Deadline: EOD      3595 Main St
Package 29 delivered at 8:29 AM      Deadline: 10:30:00 1330 2100 S
Package 31 delivered at 8:46 AM      Deadline: 10:30:00 3365 S 900 W
Package 33 delivered at 8:34 AM      Deadline: EOD      2530 S 500 E
Package 34 delivered at 8:11 AM      Deadline: 10:30:00 4580 S 2300 E
Package 39 delivered at 9:05 AM      Deadline: EOD      2010 W 500 S
Package 40 delivered at 8:40 AM      Deadline: 10:30:00 380 W 2880 S

Truck 2 16 packages (en route (departed 9:05 AM))
Package 1 en route                 Deadline: 10:30:00 195 W Oakland Ave
Package 3 en route                 Deadline: EOD      233 Canyon Rd
Package 4 en route                 Deadline: EOD      380 W 2880 S
Package 6 en route                 Deadline: 10:30:00 3060 Lester St
Package 10 en route                Deadline: EOD      600 E 900 South
Package 11 delivered at 9:29 AM      Deadline: EOD      2600 Taylorsville Blvd
Package 17 en route                 Deadline: EOD      3148 S 1100 W
Package 18 delivered at 9:33 AM      Deadline: EOD      1488 4800 S
Package 23 delivered at 9:31 AM      Deadline: EOD      5100 South 2700 West
Package 25 delivered at 9:13 AM      Deadline: 10:30:00 5383 S 900 East #104
Package 28 en route                 Deadline: EOD      2835 Main St
Package 30 en route                 Deadline: 10:30:00 300 State St
Package 32 en route                 Deadline: EOD      3365 S 900 W
Package 36 en route                 Deadline: EOD      2300 Parkway Blvd
Package 37 en route                 Deadline: 10:30:00 300 State St
Package 38 en route                 Deadline: EOD      300 State St

Truck 3 8 packages (at hub)
Package 5 at hub                   Deadline: EOD      300 State St
Package 8 at hub                   Deadline: EOD      300 State St
Package 9 delayed                   Deadline: EOD      410 S State St
Package 22 at hub                   Deadline: EOD      6351 South 900 East
Package 24 at hub                   Deadline: EOD      5025 State St
Package 26 at hub                   Deadline: EOD      5383 S 900 East #104
Package 27 at hub                   Deadline: EOD      1060 Dalton Ave S
Package 35 at hub                   Deadline: EOD      1060 Dalton Ave S

Summary
delivered: 18
en route: 14
at hub: 7
delayed: 1

Enter a time (e.g. '8:35 AM', '12:03 PM', '1:12 PM') or 'all' for final status, or 'q' to quit:
```

Status Check at 12:03 p.m.

Figure 3. Package status at 12:03 p.m. All forty packages show “delivered at “ with no pending deliveries. Truck 3’s last deliveries occurred around 11:10 a.m. to 11:46 a.m., confirming that the day ends only when every package has been served.

Enter a time (e.g. '8:35 AM', '12:03 PM', '1:12 PM') or 'all' for final status, or 'q' to quit: 12:03 PM

Status at 12:03 PM

Truck 1 16 packages (returned to hub at 9:42 AM)

Package 2	delivered at 8:34 AM	Deadline: EOD	2530 S 500 E
Package 7	delivered at 8:29 AM	Deadline: EOD	1330 2100 S
Package 12	delivered at 9:29 AM	Deadline: EOD	3575 W Valley Central Station bus Loop
Package 13	delivered at 9:05 AM	Deadline: 10:30:00	2010 W 500 S
Package 14	delivered at 8:18 AM	Deadline: 10:30:00	4300 S 1300 E
Package 15	delivered at 8:11 AM	Deadline: 09:00:00	4580 S 2300 E
Package 16	delivered at 8:11 AM	Deadline: 10:30:00	4580 S 2300 E
Package 19	delivered at 9:34 AM	Deadline: EOD	177 W Price Ave
Package 20	delivered at 9:36 AM	Deadline: 10:30:00	3595 Main St
Package 21	delivered at 9:36 AM	Deadline: EOD	3595 Main St
Package 29	delivered at 8:29 AM	Deadline: 10:30:00	1330 2100 S
Package 31	delivered at 8:46 AM	Deadline: 10:30:00	3365 S 900 W
Package 33	delivered at 8:34 AM	Deadline: EOD	2530 S 500 E
Package 34	delivered at 8:11 AM	Deadline: 10:30:00	4580 S 2300 E
Package 39	delivered at 9:05 AM	Deadline: EOD	2010 W 500 S
Package 40	delivered at 8:40 AM	Deadline: 10:30:00	380 W 2880 S

Truck 2 16 packages (returned to hub at 10:52 AM)

Package 1	delivered at 10:09 AM	Deadline: 10:30:00	195 W Oakland Ave
Package 3	delivered at 10:26 AM	Deadline: EOD	233 Canyon Rd
Package 4	delivered at 10:03 AM	Deadline: EOD	380 W 2880 S
Package 6	delivered at 9:51 AM	Deadline: 10:30:00	3060 Lester St
Package 10	delivered at 10:36 AM	Deadline: EOD	600 E 900 South
Package 11	delivered at 9:29 AM	Deadline: EOD	2600 Taylorsville Blvd
Package 17	delivered at 9:56 AM	Deadline: EOD	3148 S 1100 W
Package 18	delivered at 9:33 AM	Deadline: EOD	1488 4800 S
Package 23	delivered at 9:31 AM	Deadline: EOD	5100 South 2700 West
Package 25	delivered at 9:13 AM	Deadline: 10:30:00	5383 S 900 East #104
Package 28	delivered at 10:07 AM	Deadline: EOD	2835 Main St
Package 30	delivered at 10:24 AM	Deadline: 10:30:00	300 State St
Package 32	delivered at 9:58 AM	Deadline: EOD	3365 S 900 W
Package 36	delivered at 9:46 AM	Deadline: EOD	2300 Parkway Blvd
Package 37	delivered at 10:24 AM	Deadline: 10:30:00	300 State St
Package 38	delivered at 10:24 AM	Deadline: EOD	300 State St

Truck 3 8 packages (returned to hub at 11:58 AM)

Package 5	delivered at 11:10 AM	Deadline: EOD	300 State St
Package 8	delivered at 11:10 AM	Deadline: EOD	300 State St
Package 9	delivered at 11:13 AM	Deadline: EOD	410 S State St
Package 22	delivered at 11:46 AM	Deadline: EOD	6351 South 900 East
Package 24	delivered at 11:36 AM	Deadline: EOD	5025 State St
Package 26	delivered at 11:42 AM	Deadline: EOD	5383 S 900 East #104
Package 27	delivered at 10:54 AM	Deadline: EOD	1060 Dalton Ave S
Package 35	delivered at 10:54 AM	Deadline: EOD	1060 Dalton Ave S

Summary

delivered: 40

Enter a time (e.g. '8:35 AM', '12:03 PM', '1:12 PM') or 'all' for final status, or 'q' to quit: █

Total Mileage

Figure 4. Total mileage after complete execution. The output header reads “Total mileage for all trucks: 89.3 miles,” which is below the 140-mile requirement. Every package shows its final delivered status, truck assignment, and timestamp.

Enter a time (e.g. '8:35 AM', '12:03 PM', '1:12 PM') or 'all' for final status, or 'q' to quit: all

Truck 1 16 packages 30.7 miles departed 8:00 AM

Package	2	delivered at 8:34 AM	Deadline: EOD	2530 S 500 E
Package	7	delivered at 8:29 AM	Deadline: EOD	1330 2100 S
Package	12	delivered at 9:29 AM	Deadline: EOD	3575 W Valley Central Station bus Loop
Package	13	delivered at 9:05 AM	Deadline: 10:30:00	2010 W 500 S
Package	14	delivered at 8:18 AM	Deadline: 10:30:00	4300 S 1300 E
Package	15	delivered at 8:11 AM	Deadline: 09:00:00	4580 S 2300 E
Package	16	delivered at 8:11 AM	Deadline: 10:30:00	4580 S 2300 E
Package	19	delivered at 9:34 AM	Deadline: EOD	177 W Price Ave
Package	20	delivered at 9:36 AM	Deadline: 10:30:00	3595 Main St
Package	21	delivered at 9:36 AM	Deadline: EOD	3595 Main St
Package	29	delivered at 8:29 AM	Deadline: 10:30:00	1330 2100 S
Package	31	delivered at 8:46 AM	Deadline: 10:30:00	3365 S 900 W
Package	33	delivered at 8:34 AM	Deadline: EOD	2530 S 500 E
Package	34	delivered at 8:11 AM	Deadline: 10:30:00	4580 S 2300 E
Package	39	delivered at 9:05 AM	Deadline: EOD	2010 W 500 S
Package	40	delivered at 8:40 AM	Deadline: 10:30:00	380 W 2880 S

Truck 2 16 packages 32.2 miles departed 9:05 AM

Package	1	delivered at 10:09 AM	Deadline: 10:30:00	195 W Oakland Ave
Package	3	delivered at 10:26 AM	Deadline: EOD	233 Canyon Rd
Package	4	delivered at 10:03 AM	Deadline: EOD	380 W 2880 S
Package	6	delivered at 9:51 AM	Deadline: 10:30:00	3060 Lester St
Package	10	delivered at 10:36 AM	Deadline: EOD	600 E 900 South
Package	11	delivered at 9:29 AM	Deadline: EOD	2600 Taylorsville Blvd
Package	17	delivered at 9:56 AM	Deadline: EOD	3148 S 1100 W
Package	18	delivered at 9:33 AM	Deadline: EOD	1488 4800 S
Package	23	delivered at 9:31 AM	Deadline: EOD	5100 South 2700 West
Package	25	delivered at 9:13 AM	Deadline: 10:30:00	5383 S 900 East #104
Package	28	delivered at 10:07 AM	Deadline: EOD	2835 Main St
Package	30	delivered at 10:24 AM	Deadline: 10:30:00	300 State St
Package	32	delivered at 9:58 AM	Deadline: EOD	3365 S 900 W
Package	36	delivered at 9:46 AM	Deadline: EOD	2300 Parkway Blvd
Package	37	delivered at 10:24 AM	Deadline: 10:30:00	300 State St
Package	38	delivered at 10:24 AM	Deadline: EOD	300 State St

Truck 3 8 packages 26.4 miles departed 10:30 AM

Package	5	delivered at 11:10 AM	Deadline: EOD	300 State St
Package	8	delivered at 11:10 AM	Deadline: EOD	300 State St
Package	9	delivered at 11:13 AM	Deadline: EOD	410 S State St
Package	22	delivered at 11:46 AM	Deadline: EOD	6351 South 900 East
Package	24	delivered at 11:36 AM	Deadline: EOD	5025 State St
Package	26	delivered at 11:42 AM	Deadline: EOD	5383 S 900 East #104
Package	27	delivered at 10:54 AM	Deadline: EOD	1060 Dalton Ave S
Package	35	delivered at 10:54 AM	Deadline: EOD	1060 Dalton Ave S

Summary

delivered: 40
Total mileage: 89.3 miles

Enter a time (e.g. '8:35 AM', '12:03 PM', '1:12 PM') or 'all' for final status, or 'q' to quit: